Malus' Law

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Aim of the Experiment

To verify Malus' Law of polarization of light.

Apparatus Required

(i) Optical bench.
(ii) Laser light source.
(iii) Polarizer with stem for mounting.
(iv) Photometer (Optical base plate).
(v) Digital multimeter with connecting wire.

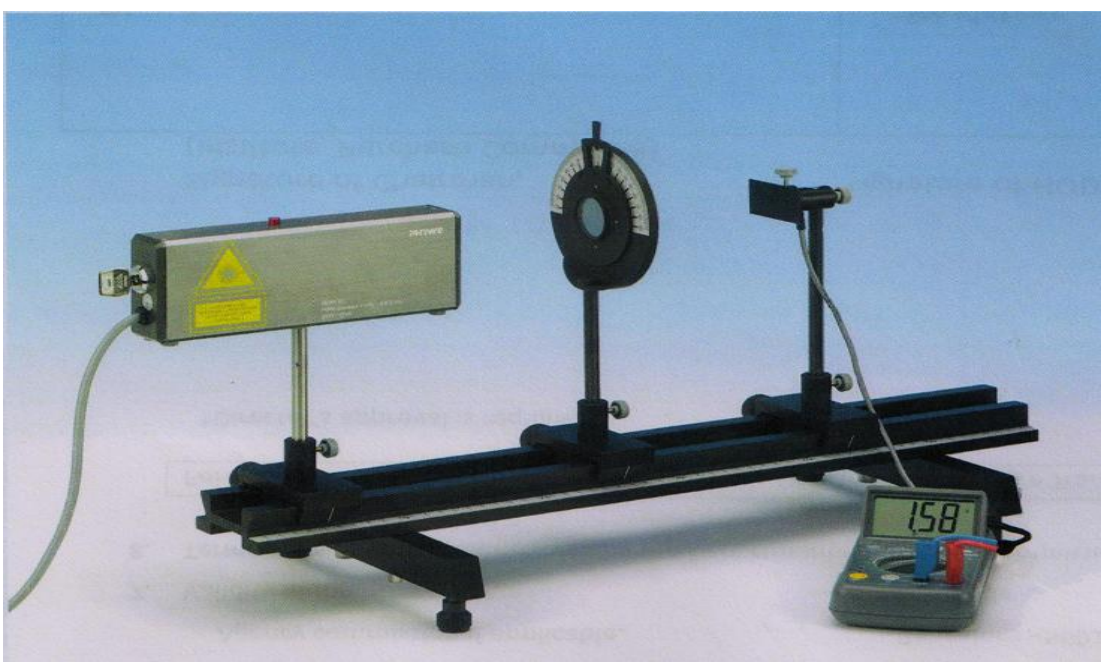


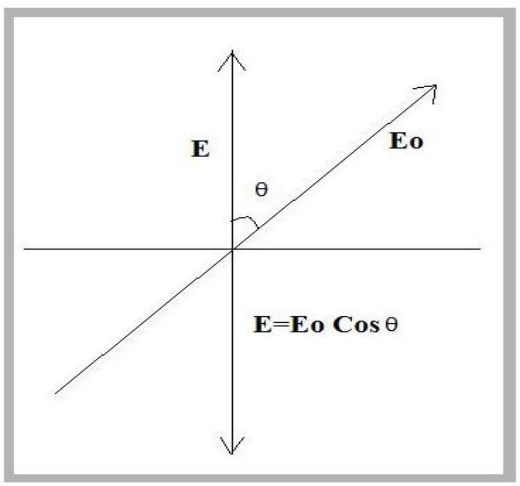
Fig-1: Experimental setup

Theory

Figure-1 depicts the complete experimental set up for verification of Malus' law of polarization. When linearly polarized light passes through a Polaroid, the intensity of the transmitted light varies with the angular position of the Polaroid w.r.t. the angle of the plane of polarization of incident light. The emergent light after falling on the photo detector, gives a photocurrent which is proportional to the intensity of the emergent light.

Mathematically

$$I = I_0 \cos^2 \theta$$



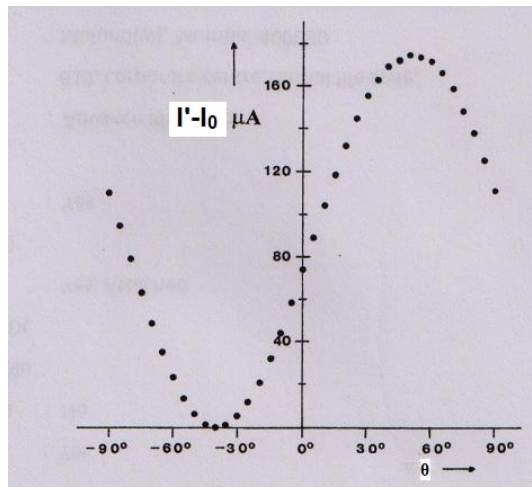


Fig-2(a): Fig-2(b)

Where, I is the photocurrent for the emergent light.

I_0 is the photocurrent for the incident light

'θ' corresponds to the angle between the direction of polarization of the incident light and the axis of the Polaroid.

This law can be illustrated in the form of amplitude of incident polarized light (E_0) and emergent light (E) as

$$E = E_0 \cos\theta$$

where E and E_0 are the electric vector associated with the emergent and incident light shown in Fig-2(a). So if we rotate the Polaroid in the clockwise manner, at a

particular angle the intensity of the emergent light will be maximum giving highest value of photo current and if we continue in the same direction, finally the intensity will decrease to almost zero illustrated. The graph between photocell current as a function of angular position θ of the plane of polarization of the analyzer is illustrated in Fig-2(b).

Procedure

(i) Place the set-up is in a dark room so that excess light except the laser source may be avoided.

(ii) Adjusted the source of light, Polaroid and the photometer on the optical bench so that the light after passing through the Polaroid falls at the centre of the photometer.

(iii) Before the laser is switched on note down the effect of the stray light on the photometer which will be finally deducted from the observed values.

(iv) Switch on the laser source for 30 minutes and record the signal from the digital multi-meter for each specific positions of the Polaroid.

(v) Plot graph between the square of the cosine of the polarising angle vs. the photocurrent (I) from 0^0 to 90^0 and back to 0^0.

(vi) Plot the graph between the differential current $I'-I_0$ and θ.

Observations

Table-1: Table for current

Sl. No.	Stray current $I_S(\mu A)$	θ	$Cos^2\theta$	Photocurrent $I(\mu A)$	$I-I_S(\mu A)$ =I'	Difference Current $I'-I_0(\mu A)$
1.	0	+90	0	6.8	6.8	-121.4
2.	0	+85	7.5961×10^{-3}	7.2	7.2	-121
3.	0	+80	30.1536×10^{-3}	8.4	8.4	-119.8
4.	0	+75	66.9872×10^{-3}	10.8	10.8	-117.4
5.	0	+70	116.977×10^{-3}	15.4	15.4	-112.8
6.	0	+65	178.606×10^{-3}	21.2	21.2	-107

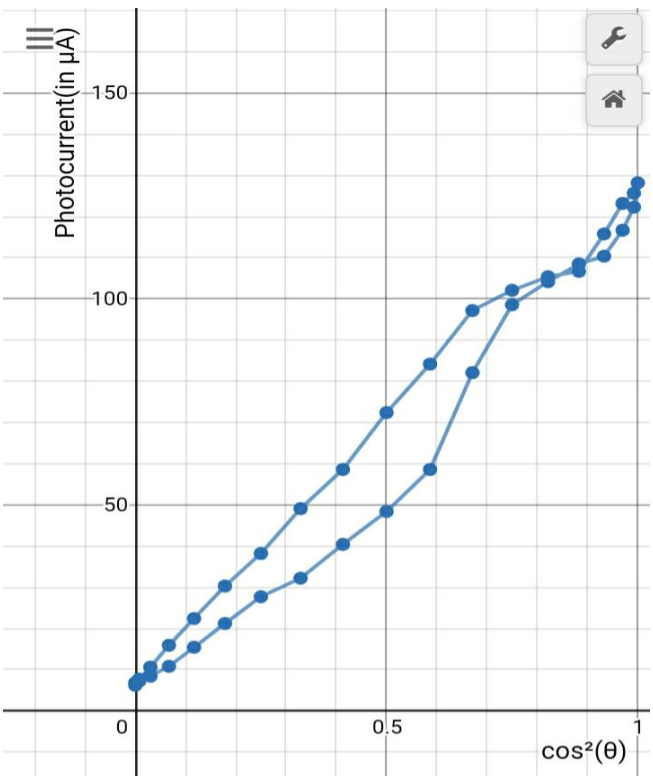
Sl. No.	Stray current $I_S(\mu A)$	θ	$Cos^2\theta$	Photocurrent $I(\mu A)$	$I-I_S(\mu A)$ =I'	Difference Current $I'-I_0(\mu A)$
7.	0	+60	250×10^{-3}	27.7	27.7	-100.5
8.	0	+55	328.98×10^{-3}	32.2	32.2	-96
9.	0	+50	413.176×10^{-3}	40.4	40.4	-87.8
10.	0	+45	500×10^{-3}	48.4	48.4	-79.8
11.	0	+40	586.824×10^{-3}	58.6	58.6	-69.6
12.	0	+35	671.01×10^{-3}	82.1	82.1	-46.1
13	0	+30	750×10^{-3}	96.8	98.6	-29.6
14.	0	+25	821.394×10^{-3}	104.2	104.2	-24
15.	0	+20	883.022×10^{-3}	108.5	108.5	-19.7

16.	0	+15	933.013×10^{-3}	110.4	110.4	-17.8
17.	0	+10	969.846×10^{-3}	116.7	116.7	-11.5
18.	0	+5	992.404×10^{-3}	122.3	122.3	-5.9
19.	0	0	1000×10^{-3}	128.2	128.2	0
20.	0	-5	992.404×10^{-3}	125.7	125.7	-2.5
21.	0	-10	969.846×10^{-3}	123.2	123.2	-5
22.	0	-15	933.013×10^{-3}	115.8	115.8	-12.4
23.	0	-20	883.022×10^{-3}	106.7	106.7	-21.5
24.	0	-25	829.394×10^{-3}	105.4	105.4	-22.8
25.	0	-30	750×10^{-3}	102.1	102.1	-26.1
26.	0	-35	671.01×10^{-3}	97.2	97.2	-31
27.	0	-40	586.824×10^{-3}	84.2	84.2	-44
28.	0	-45	500×10^{-3}	72.4	72.4	-55.8

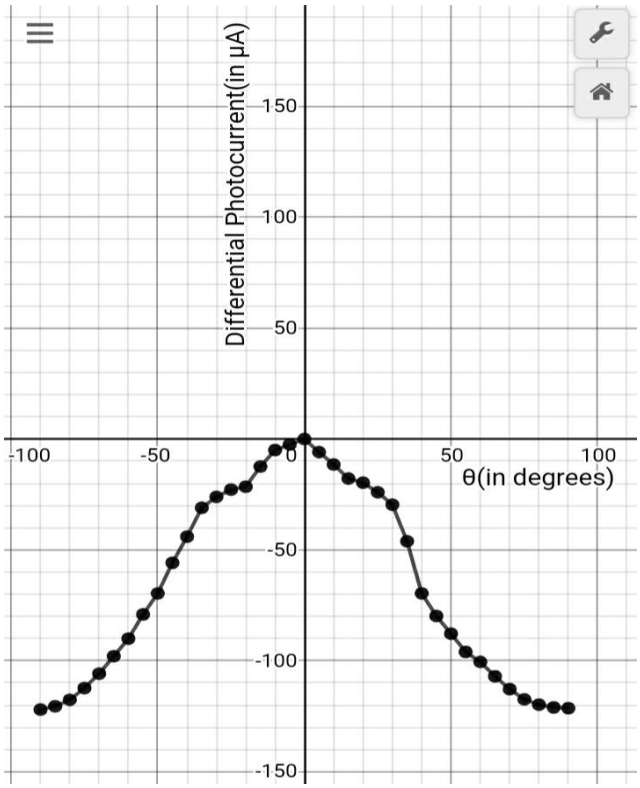
Sl. No.	Stray current I_S (μA)	θ	$Cos^2\theta$	Photocurrent I(μA)	$I-I_S$(μA) =I'	Difference Current $I'-I_0$(μA)
29.	0	-50	413.176×10^{-3}	58.6	58.6	-69.6
30.	0	-55	328.98×10^{-3}	49.1	49.1	-79.1
31.	0	-60	250×10^{-3}	38.2	38.2	-90
32.	0	-65	178.606×10^{-3}	30.3	30.3	-97.9
33.	0	-70	116.977×10^{-3}	22.4	22.4	-105.8
34.	0	-75	66.9872×10^{-3}	15.9	15.9	-112.3

35.	0	-80	30.1536×10^{-3}	10.6	10.6	-117.6
36.	0	-85	7.5961×10^{-3}	7.7	7.7	-120.5
37.	0	-90	0	6.2	6.2	-122

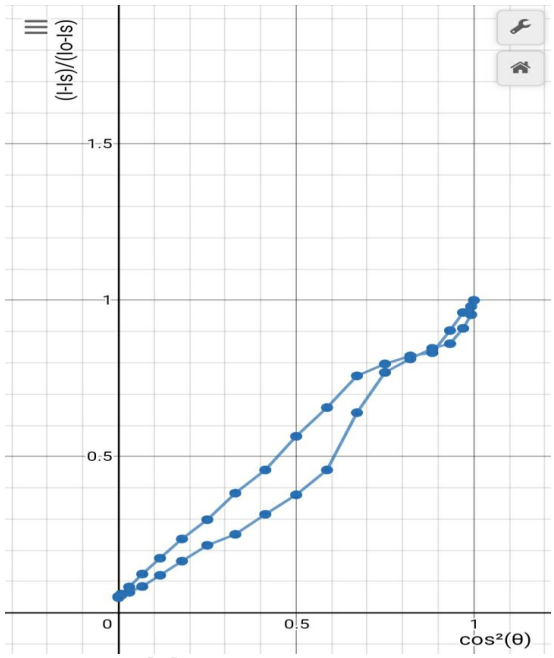
Graphs and Results



Photocurrent(I) vs. $\cos^2\theta$.



Differential Photocurrent($I'-I_0$)vs θ



$\frac{I-Is}{I_0-Is}$ **vs.** $\cos^2\theta$

Result

In **photocurrent(I) vs. cos²θ** , expected graph was a straight line which the experimental plot closely resembles. In **differential Photocurrent($I'-I_0$)vs θ** graph, the expected plot was a -sin²θ plot which is confirmed. In $\frac{I-Is}{I_0-Is}$ **vs.** $\cos^2\theta$graph, the expected plot was a straight line with slope 1 which closely resembles the plot (neglecting small deviations). Thus the validity of malus's law is experimentally verified.

Precautions

(i) One should not look directly to LASER light because it may damage the eye.

(ii) The optical bench should be levelled properly.

(iii) One should carefully handle the polarizer because slight disturbance may disturb the result obtained.

Considerations

(i) What is polarization of light and how the light gets polarized?

Ans: Polarization is a property of transverse waves that characterizes the geometric orientation of vibrations of a particular quantity of interest(like displacement for waves on a string and electric field for EM waves). Light may be polarized by using polaroid filters that only allow that component of light to pass through which has axis of electric field oscillations oriented a particular way with respect to the filter and the other components are absorbed. The light thus produced is called plane polarized light.

(ii) What is plane of polarization and plane of vibration?

Ans: The plane to which the vibration of electric field and magnetic field of light is confined is said to be the plane of vibration and the plane of polarization is the plane that contains the electric field vector and the direction of propagation.

(iii) What is Brewster's law and what are polarizer and analyzer?

Ans: Brewster's angle (also known as the polarization angle) is an angle of incidence at which light with a particular polarization is perfectly transmitted through a transparent dielectric surface, with no reflection. When unpolarized light is incident at this angle, the light that is reflected from the surface is therefore perfectly polarized. Polarizers and analyzers are parts of optical instruments that use plane polarized light. There are many types of polarizers and analyzers that can be chosen according to our needs. A polarizer can filter light waves in order to generate polarization of light. In other words, a polarizer can generate plane polarized light from light waves coming from a normal light source. The analyzer acts as a second polarizer. The main difference between polarizer and analyzer is that polarizer produces plane polarized light whereas analyzer can be used to check whether the light has been polarized or not.

(iv) How circularly, elliptically and linearly polarized light are produced and what are their properties?

Ans: Circular polarization of an electromagnetic wave is a polarization state in which, at each point, the electromagnetic field of the wave has a constant magnitude and is rotating at a constant rate in a plane perpendicular to the direction of the wave. Circular polarization occurs when the two orthogonal electric field component vectors are of equal magnitude and are out of phase by exactly 90°, or one-quarter wavelength. Elliptical polarization occurs when the magnitudes of constituent components are not same or the phase difference is not exactly 90°.

Conclusions

(i) In this experiment we confirm the fact that light is a transverse electromagnetic wave and can be polarized and obeys malus's law. The plots in the experiment do not completely match with the theoretical graphs due to unforeseen human error, imperfect polarizers and background lighting. Nevertheless the resemblance shared by the two is excellent for making a conclusion.

(ii) The experiment can be made more accurate by taking into consideration background lighting, and using good quality polarizers.

(iii) Polarizers and analysers are ubiquitous objects in modern science. They are used in a wide range of applications like photography, chemistry, biology, study of crystals and optically active compounds like drugs and proteins. The seemingly innocent idea of light waves interacting with different media gives rise to a plethora of interesting phenomenon which makes polarization an important tool with which we can study the properties of the media it interacts with.